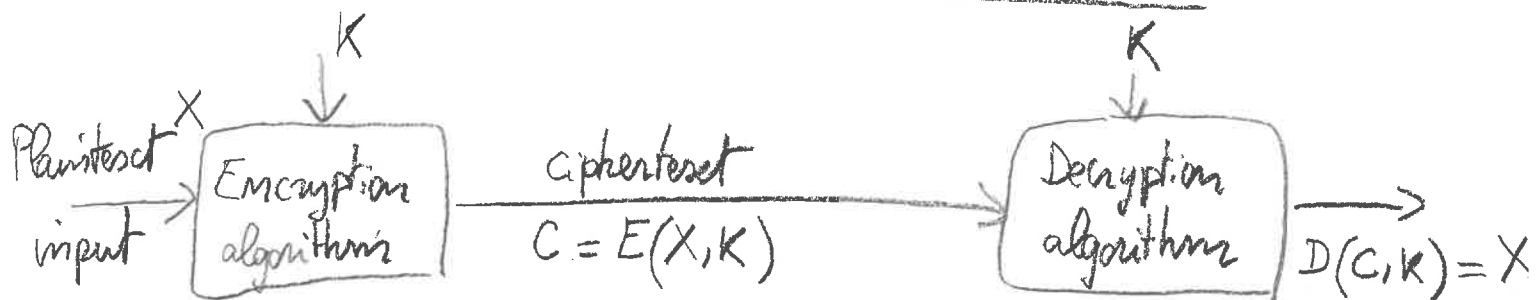


Chapter 2

Symmetric Cryptosystems



K is a secret key shared by the sender and recipient
 K is independent of the plaintext X and of the encryption and decryption algorithms

Requirements

- The key must be secret and only known by the sender and recipient
- It should be impractical to recover X when observing C

Kerckhoff's principle:

We should not need to keep the encryption/decryption algorithms secret. Only the key should be kept secret.

- Why?
- Easier to change keys than to change algorithms
 - Manufacturers can develop large-scale chip implementations of encryption algorithms

- Efficient in multiuser systems.
Easy to share algorithm

Examples of attacks:

Brute-force attack: Try every possible key
Very easy to do but maybe impractical
(take too long)

Known plaintext attack: The adversary knows
 X' and $C' = E(X', K)$
and wants to decipher $C = E(X, K)$

I Examples of symmetric cryptosystems

A. Shift and Caesar cipher

Used by Julius Caesar to encrypt his communication

Idea: Shift the letters of the alphabet by three positions

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Plaintext: JULIUS

Ciphertext: MXOLXV

Ciphertext: FDHVDU

Plaintext: CAESAR

Review: modulo operations

- $13 \bmod 3 = 1$

$$13 \equiv 1 \pmod{3}$$

- $9 \bmod 3 = 0$

- $-3 \bmod 26 = 23$

- $-15 \bmod 7 = 6$

$$\left\{ \begin{array}{l} 13 = 3 \times 4 + 1 \\ 13 - 3 = 10 \\ 10 - 3 = 7 \\ 7 - 3 = 4 \\ 4 - 3 = 1 \end{array} \right.$$

$$\left\{ \begin{array}{l} -3 + 26 = 23 \end{array} \right.$$

$$\left\{ \begin{array}{l} -15 + 7 = -8 \\ -8 + 7 = -1 \\ -1 + 7 = 6 \end{array} \right.$$

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

X = plaintext is one letter ($\equiv$ number between 0 and 25)

$$C = E(X, K) = X + K \pmod{26}$$

||
3

$$X = D(C, K) = C - K \pmod{26}$$

Is it a good cryptosystem?

Easy to break with a brute-force attack

B. Substitution cipher

Review permutation

What are the permutations of the set $\{a, b, c\}$?

abc, cba, acb, cab, bca, bac

Idea of substitution cipher: instead of shifting the alphabet, do a permutation of the alphabet

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
E F A Q H V B N P O G L U T M X D S Y R J I Z W K C

Encryption example: (same as Caesar cipher)

Plaintext: Caesar $\rightarrow$ Cyphertext: AEHYES

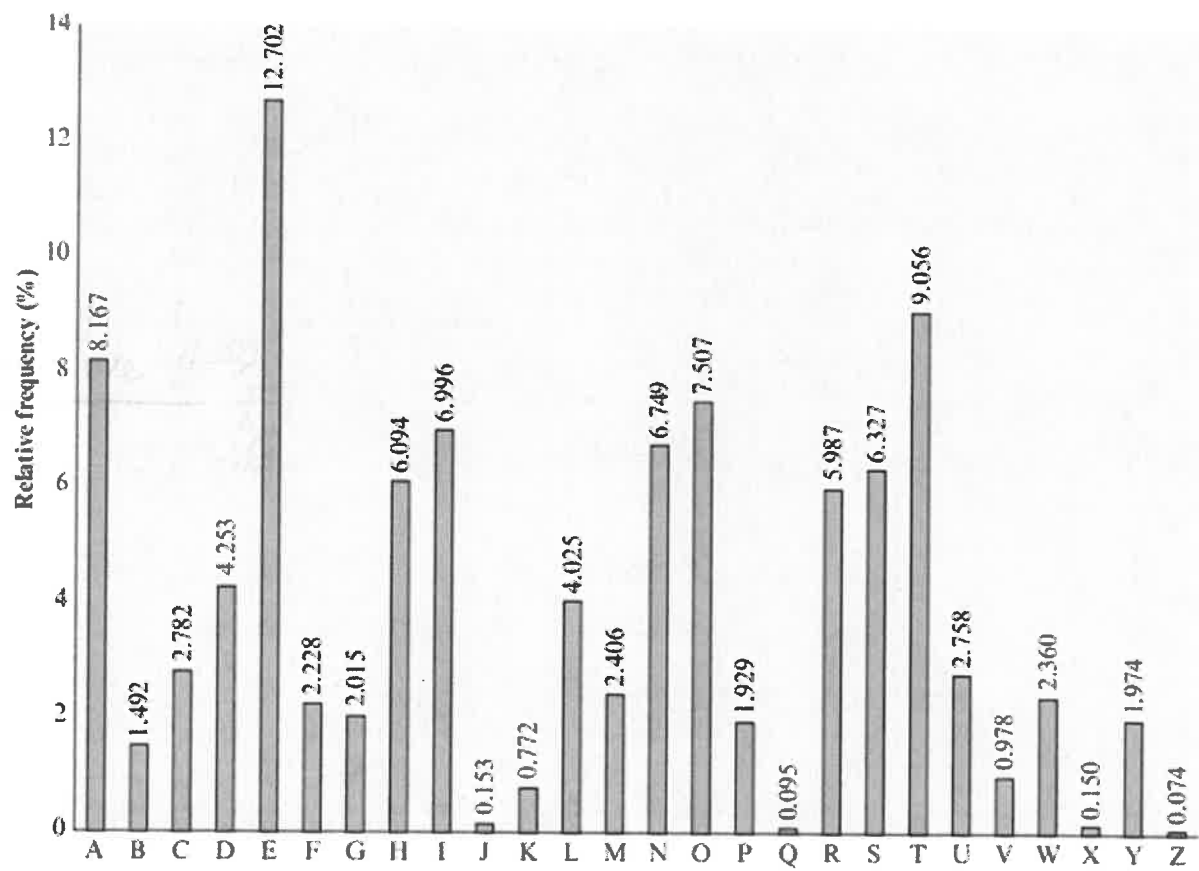
$\hookrightarrow$ The key is the permutation of the alphabet

$\hookrightarrow$ Brute-force attack: worst case number of tries $26!$
 $\approx 4 \cdot 10^{26}$

Is it a good cryptosystem?

A frequency analysis of the number of appearances of each letter in the cyphertext could reveal part of the key.
Works well if the cyphertext is long.

For short cyphertext, a frequency analysis could also work but is more involved (for instance, looking at frequencies of pairs of letters)



Frequency of Character Pairs in English Language Text

Row x , column y of the table below gives an estimate of the relative frequency of the two-character sequence xy in English language text. Specifically, it estimates the number of occurrences of xy per 10,000 characters of text. The estimate applies to English language text with characters other than letters

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	1	20	33	52	0	12	18	5	39	1	12	57	26	181	1	20	1	75	95	104	9	20	13	1	26	1
b	11	1	0	0	47	0	0	0	6	1	0	17	0	0	19	0	0	11	2	1	21	0	0	0	11	0
c	31	0	4	0	38	0	0	38	10	0	18	9	0	0	45	0	1	11	1	15	7	0	0	0	1	0
d	48	20	9	13	57	11	7	25	50	3	1	11	14	16	41	6	0	14	35	56	10	2	19	0	10	0
e	110	23	45	126	48	30	15	33	41	3	5	55	47	111	33	28	2	169	115	83	6	24	50	9	26	0
f	25	2	3	2	20	11	1	8	23	1	0	8	5	1	40	2	0	16	5	37	8	0	3	0	2	0
g	24	3	2	2	28	3	4	35	18	1	0	7	3	4	23	1	0	12	9	16	7	0	5	0	1	0
h	114	2	2	1	302	2	1	6	97	0	0	2	3	1	49	1	0	8	5	32	8	0	4	0	4	0
i	10	5	32	33	23	17	25	6	1	1	8	37	37	179	24	6	0	27	86	93	1	14	7	2	0	2
j	2	0	0	0	2	0	0	0	3	0	0	0	0	0	3	0	0	0	0	0	8	0	0	0	0	0
k	6	1	1	1	29	1	0	2	14	0	0	2	1	9	4	0	0	0	5	4	1	0	2	0	2	0
l	40	3	2	36	64	10	1	4	47	0	3	56	4	2	41	3	0	2	11	15	8	3	5	0	31	0
m	44	7	1	1	68	2	1	3	25	0	0	1	5	2	29	11	0	3	10	9	8	0	4	0	18	0
n	40	7	25	146	66	8	92	16	33	2	8	9	7	8	60	4	1	3	33	106	6	2	12	0	11	0
o	16	12	13	18	5	80	7	11	12	1	13	26	48	106	36	15	0	84	28	57	115	12	46	0	5	1
p	23	1	0	0	30	1	0	3	12	0	0	15	1	0	21	10	0	18	5	11	6	0	1	0	1	0
q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0
r	50	7	10	20	133	8	10	12	50	1	8	10	14	16	55	6	0	14	37	42	12	4	11	0	21	0
s	67	11	17	7	74	11	4	50	49	2	6	13	12	10	57	20	2	4	43	109	20	2	24	0	4	0
t	59	10	11	7	75	9	3	330	76	1	2	17	11	7	115	4	0	28	34	56	17	1	31	0	16	0
u	7	5	12	7	7	2	14	2	8	0	1	34	8	36	1	16	0	44	35	48	0	0	2	0	1	0
v	5	0	0	0	65	0	0	0	11	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	1	0
w	66	1	1	2	39	1	0	44	39	0	0	2	1	12	29	0	0	3	4	4	1	0	2	0	1	0
x	1	0	2	0	1	0	0	0	2	0	0	0	0	0	0	3	0	0	0	3	0	0	0	0	0	0
y	18	7	6	6	14	7	3	10	11	1	1	4	6	3	36	4	0	3	19	20	1	1	12	0	2	0
z	1	0	0	0	3	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

[illegible]